Results:

(a) median_age = 24.0

(b) 437 data entries

(c)

We calculated the average fare price grouped by survival status:

- Passengers who did not survive (Survived = 0): Average Fare = 22.12 (549 passengers)

- Passengers who survived (Survived = 1): Average Fare = 48.40 (342 passengers)

The average fare of survivors is more than twice as high as that of non-survivors.

Therefore, the hypothesis that "the average fare price of survivors is not very different from that of non-survivors" is not supported by the data.

Solutions:

(a)

1. Implement our own UDAF (User-Defined Aggregate Function).

2. Fill in all the blanks(null) with the age: 23

3. Sort all ages from youngest to oldest, and then find the middle age.

SELECT MedianAge(Age) FROM titanic

(b)

Since you have already registered the MedianAge(age) UDF in (a), we can use it directly in (b). Using SQL :

SELECT name, fare FROM titanic WHERE age > MedianAge(age)

(c)

1. Calculate two averages:

Average fare for survivors (Survived = 1) → avg_fare_survived

Average fare for non-survivors (Survived = 0) → avg_fare_not_survived

2. Display these two average values for a simple comparison.

3. The ticket price for survivors is more than twice that of non-survivors, a significant difference.

AI platform: Grok 4.20

The most important prompt:

(Following the titanic.csv and DataFrame.scala), Please provide the solution to problem a.

Are you using Spark SQL?

What is the difference between UDF and UADF?

Please save the results to a CSV file in the current directory.